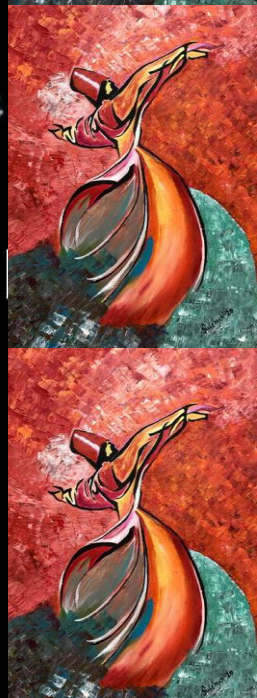




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



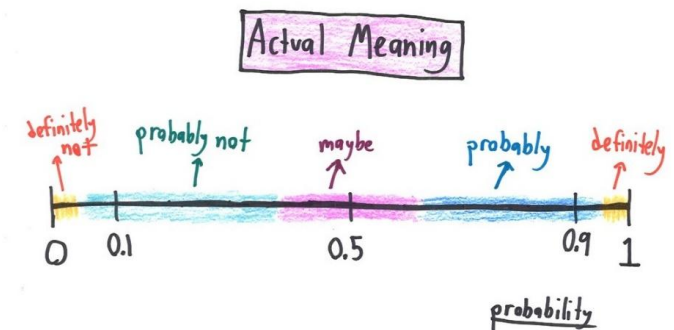


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 6 - Lecture No 11 – 24th September 2025



Announcements!

1. Lectures up to No 10 posted
2. Midterm Exam - October 8, 2025, from 05:40 PM to 06:55 PM – venue H-505(L2) – Good Luck

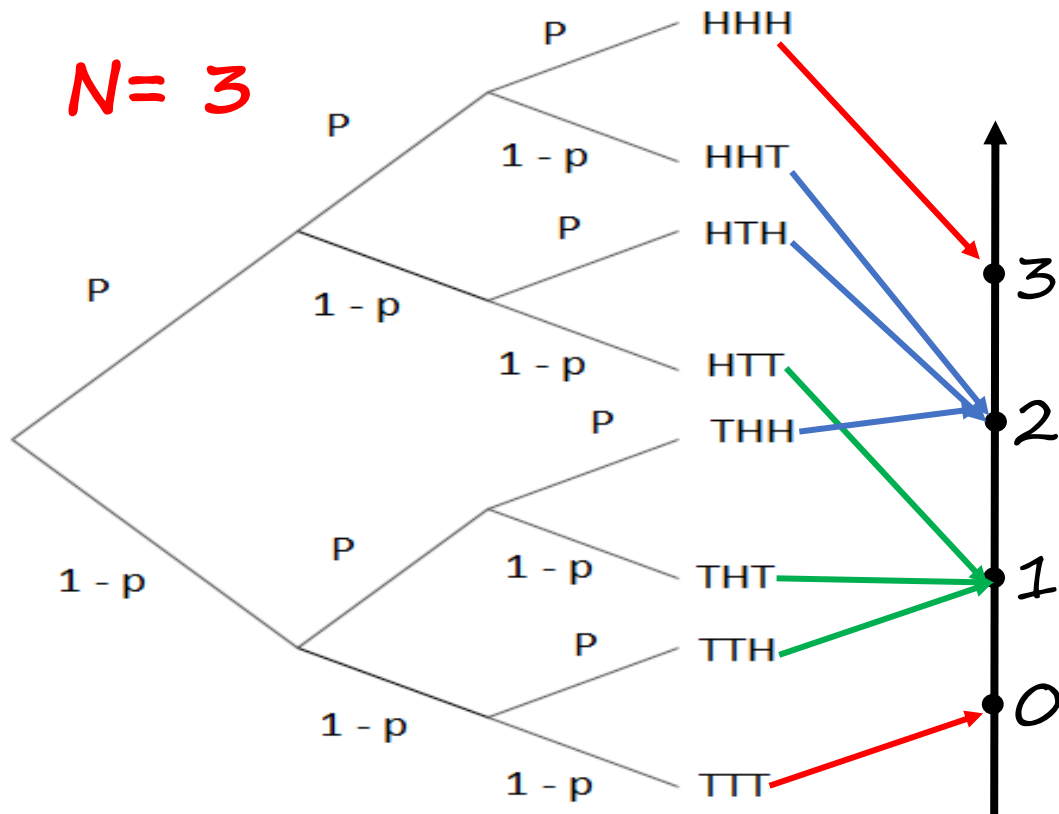
Agenda for today

1. Quiz No 03
2. Unit 4 - Discrete Random Variables

Unit 4
Discrete Random Variables

Binomial Random variable – parameters: positive integer n , $p \in [0, 1]$

- **Experiment:** n independent tosses of a coin with $P(\text{Heads}) = p$
- **Sample Space:** Set of sequences of H and T, of length n
- **Random Variable X :** number of Heads observed
- **Model of:** number of successes in a given number of independent trials



$$P_X(2) = P(X = 2)$$



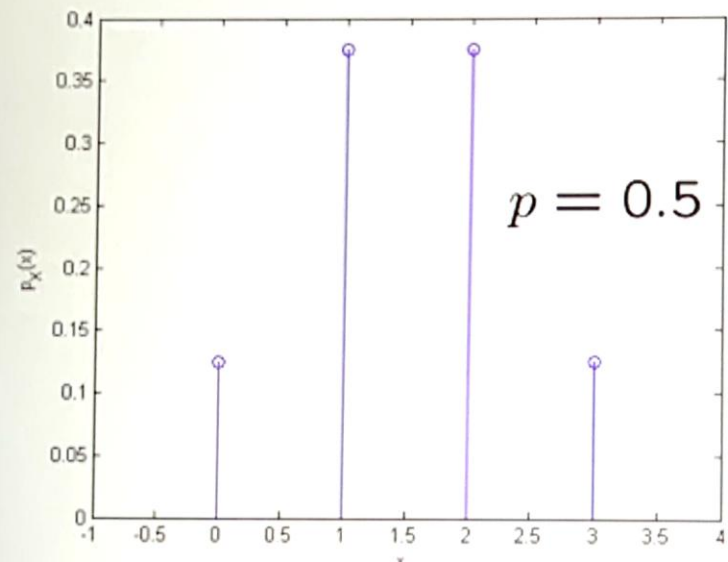
$$= P(\text{HHT}) + P(\text{HTH}) + P(\text{THH})$$



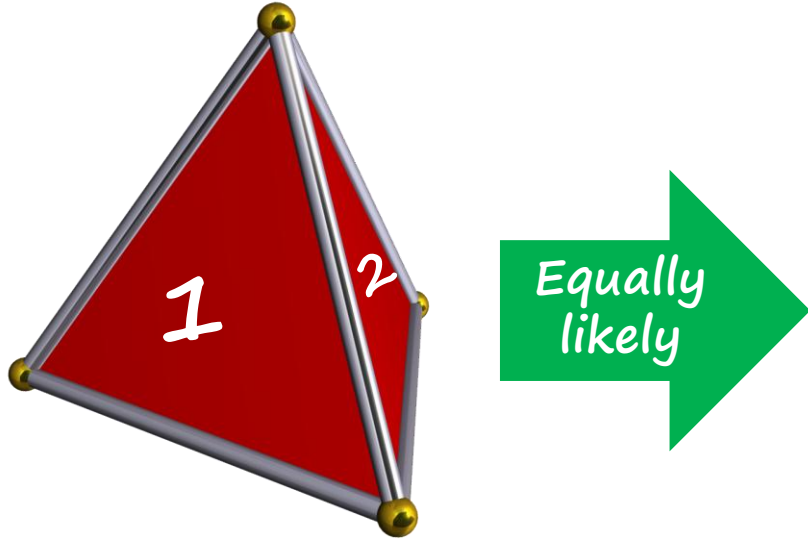
$$= 3p^2(1-p) = \binom{3}{2} p^2(1-p)$$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k = 0, 1, \dots, n$$

$$n = 3$$

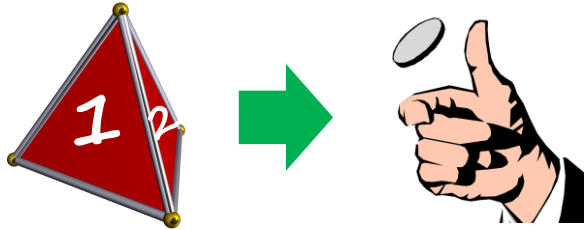


Exercise No 01!



You throw a tetrahedral dice 12 times.
R.V X is the number of times you get 3 in the 12 throws & $P_X(k)$ is the Probability of getting No 3, k -times in 12 throws

What is $P_X(5) =$



parameters

$$p = 1/4$$

$$p_X(5) = \binom{12}{5} \cdot \left(\frac{1}{4}\right)^5 \cdot \left(\frac{3}{4}\right)^7$$

Geometric Random variable – parameter: p , $0 < p \leq 1$

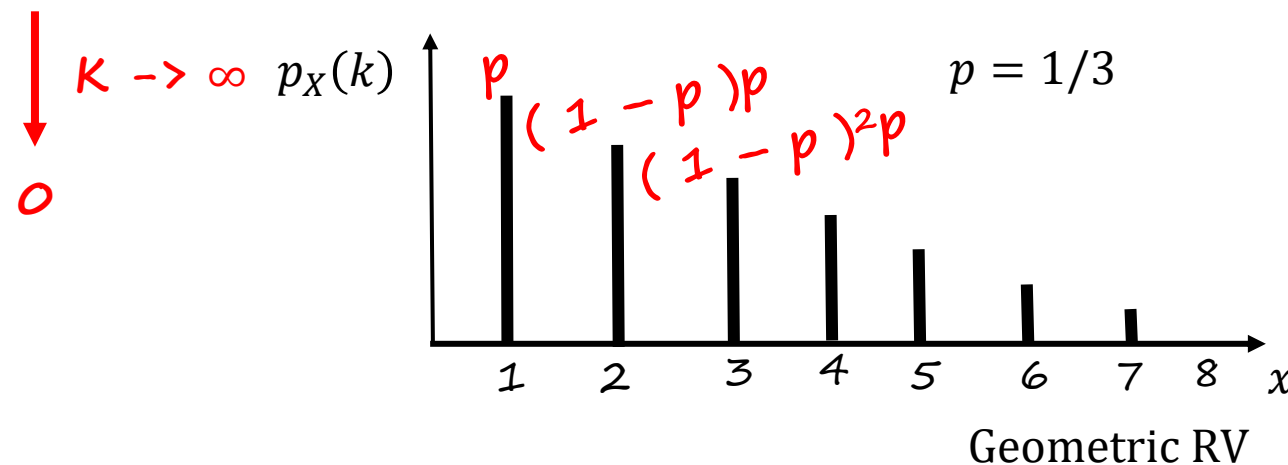
- **Experiment:** infinite many independent tosses of a coin with $P(\text{Heads}) = p$
- **Sample Space:** Set of infinite sequences of H and T
- **Random Variable X :** number of tosses until the first Heads
- **Model of:** waiting times; number of trials until a success

TTTTHTT....
X = 5

$$\rightarrow p_X(x) = P(X = k) = P(T \dots T H) = (1 - p)^{k-1} p \quad k = 1, 2, 3, \dots$$

$$\rightarrow P(\text{no heads ever}) \leq P(T \dots T) = (1 - p)^k$$

TTTTT....
"X = ∞ "



Exercise No 02!

A graduating CS student of HU goes to career fair booths in technology sector - & his/her likelihood of receiving an off-campus interview invitation after a career fair booth visit depends on how well he/she did in EE 354 course. Specifically, an A in the course results in a probability $p_1 = 0.95$ of obtaining an invitation, whereas a C in EE 354 course results in a probability of $p_2 = 0.15$ of an invitation. Assuming that each student visits 5 booths during a typical career fair, find the probability that an A grade student in EE 354 course will not get an off-campus interview invitation. Similarly, find the probability that a C grade student in EE 354 course will get an invitation during a typical career fair



$$P(\text{an A grade student does not get an invitation in 5 trials}) = \sum_{k=0}^5 (1 - p_1)^k \cdot p_1 = (1 - p_1)^5$$

$$P(\text{C grade student gets an invitation in 5 trials}) = 1 - (1 - p_2)^5$$